1. Let $A = \{a, b, .., z\}, B = \{0, 1, ..9\}$ and $C = \{0, 1, 2, 3, 4\}$ with these relations:

- R on $A \times B$ with $R(a, b)$ if a is the first letter of the English name of digit b and
- S on $B \times C$ with $S(b, c)$ if c is the remainder of $b \bmod 5$.

What are S^{-1} and $S \circ R$?

$S^{-1} = S(c, b)$, where b is the number which leaves remainder c after applying $\bmod 5$.

$S \circ R = A \times C$

2. Let $R, S, T \subseteq A \times A$ be relations on the same set A. Prove or disprove each of the following statements. If the equality does not always hold, give a concrete counterexample.

   a. $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$

   Let $(a, c) \in (R \circ S)^{-1}$

   $(c, a) \in R \circ S$ *via inverse*

   $(c, b) \in S, (b, a) \in R$ *via composition*

   $(b, c) \in S^{-1}, (a, b) \in R^{-1}$ *via inverse*

   $(a, c) \in S^{-1} \circ R^{-1}$ *via composition*

   $\left((R \circ S)^{-1} \subseteq S^{-1} \circ R^{-1}\right) \wedge \left(S^{-1} \circ R^{-1} \subseteq (R \circ S)^{-1}\right)$

   $\top$

   b. $\left(R^{-1}\right)^{-1} = R$

   Let $(a, b) \in \left(R^{-1}\right)^{-1}$

   $(b, a) \in R^{-1}$ *via inverse*

   $(a, b) \in R$ *via inverse*

   $\left(\left(R^{-1}\right)^{-1} \subseteq R\right) \wedge \left(R \subseteq \left(R^{-1}\right)^{-1}\right)$

   $\top$

   c. If $R \subseteq S$, then $R \circ T \subseteq S \circ T$

   Let $(a, c) \in R \circ T$

   $(a, b) \in T, (b, c) \in R$ *via composition*

   $(b, c) \in S$ *via subset*

   $(a, b) \in T, (b, c) \in S$

   $(a, c) \in S \circ T$ *via composition*

   $R \circ T \subseteq S \circ T$

   $\top$

d. $R \circ (S \cup T) = (R \circ S) \cup (R \circ T)$

Let $(a, c) \in R \circ (S \cup T)$

$(a, b) \in S \cup T, (b, c) \in R$ *via composition*

$((a, b) \in S) \vee ((a, b) \in T), (b, c) \in R$ *via union*

$(a, c) \in R \circ S \vee (a, c) \in R \circ T$ *via composition*

$(R \circ S) \cup (R \circ T)$ *via union*

$(R \circ (S \cup T) \subseteq (R \circ S) \cup (R \circ T)) \wedge ((R \circ S) \cup (R \circ T)) \subseteq (R \circ (S \cup T))$

$\top$

e. $R \circ (S \cap T) = (R \circ S) \cap (R \circ T)$

Let $(a, c) \in R \circ (S \cap T)$

$(a, b) \in S \cap T, (b, c) \in R$ *via composition*

$((a, b) \in S \wedge (a, b) \in T), (b, c) \in R$ *via intersection*

$(a, c) \in R \circ S \wedge (a, c) \in R \circ T$ *via composition*

$(R \circ T) \cap (R \circ S)$ *via intersection*

3. Give an example of a partial order that is not total.

The subset operator ($\subseteq$) is a partial order as it satisfies:

- Reflexivity: e.g. $a \subseteq a$
- Antisymmetry: e.g If $a \subseteq b \wedge b \subseteq a$, then $a = b$
- Transitivity: e.g. If $a \subseteq b \wedge b \subseteq c$, then $a \subseteq c$

However, it is not a total order:

Totality is defined as: $\forall a, b : S.\ (aRb \vee bRa)$

Let $a = \{1\}, b = \{2\}$, with the relation $R = \subseteq$

$a \not\subseteq b \wedge b \not\subseteq a$, therefore subset is not total.

4. On the set of integers, the relation | pronounced 'divides' is defined as follows: $a|b$ if there is some integer k, such that $ak = b$. Is this reflexible, symmetric, antisymmetric, transitive, a partial order, an equivalence?

- Reflexivity ✓: $a|a$, e.g. $2|2$
- Symmetric ✗: If $a|b$, $b|a$ does not necessarily hold. e.g. $2|4$ does not imply $4|2$
- Antisymmetric ✗: If $a|b \wedge b|a$, then $a = b$. e.g. $2|2$ implies $2 = 2$
- Transitive ✓: If $a|b \wedge b|c$, then $a|c$ since c is just a multiple of a. e.g. $2|4 \wedge 4|8$, and $2|8$
- Partial order ✗: It's not antisymmetric, thus it is not partial order.
- Equivalence ✗: It's not symmetric, thus it is not equivalent.

5. Suppose $(A, (\leq)_A)$ and $(B, (\leq)_B)$ are partially ordered sets. Prove that the product order is a partial order, but not necessarily total.

$(a_1, b_1) \leq (a_2, b_2) \iff (a_1 (\leq)_A\ a_2) \land (b_1 (\leq)_B\ b_2)$

- Reflexivity ✓. e.g. $(1,2) \leq (2,3), (1 \leq 2) \land (2 \leq 3)$
- Antisymmetric ✓. e.g. $(0,1) \leq (0,1), (0 \leq 0) \land (1 \leq 1), (0,1) = (0,1)$
- Transitive ✓. e.g. $(0,1) \leq (1,2) \land (1,2) \leq (2,3) \Rightarrow (0,1) \leq (2,3), (0 \leq 2) \land (1 \leq 3)$
- Totality ✗. e.g. $(1,2) \not\leq (2,1) \land (2,1) \not\leq (1,2)$

6.

The ancient Chinese book *I Ching* contains eight trigrams.

A broken line is called a *yin* line, and a solid line is called a *yang* line.

It is said in the West that binary numbers are partly the invention of the German Philosopher and Mathematician (among many other things) Gottfried Wilhelm Leibniz. However, Leibniz was definitely aware of the I Ching and even came up with the analogy **yin=0**, **yang=1**, so the invention of binary might be better credited to China some time around 1000 B.C

a. The relation R on the set of eight trigrams is defined as follows: $R(A, B)$ if in all the places where you put ink to draw A, you also put ink to draw B; equivalently, you could transform A into B by possibly adding ink, but not removing any. For example, $R(\text{water}, \text{lake})$ holds because you can add ink to complete the lowest line, but $\neg R(\text{lake}, \text{earth})$ because once you have drawn 'lake', you cannot make 'earth' without erasing anything.

What kind of relation is R? Prove your answer.

- Reflexivity ✓: $R(\text{heaven}, \text{heaven})$
- Symmetric ✗: $R(\text{lake}, \text{heaven}) \neq R(\text{heaven}, \text{lake})$
- Antisymmetric ✓: $aRb \land bRa \Rightarrow a = b$
- Transitive ✓: $R(\text{earth}, \text{mountain}) \land R(\text{mountain}, \text{wind}) \Rightarrow R(\text{earth}, \text{wind})$
- Partial ✓: Reflexivity, antisymmetric and transitive.
- Totality ✗: $\neg(R(\text{lake}, \text{earth}) \land R(\text{earth}, \text{lake}))$

b. The relation S on the set of eight trigrams is defined as follows: $S(A, B)$ if you can change A into B by changing at most one line (that is changing a *yin* into a *yang* line, or the other way around).

What kind of relation is S? Which properties of relations does it have?

- Reflexivity ✓: $S(\text{heaven}, \text{heaven})$
- Symmetric ✓: $S(\text{earth}, \text{water}) \land S(\text{water}, \text{earth})$
- Antisymmetric ✗: $S(\text{earth}, \text{water}) \land S(\text{water}, \text{earth}) \land \text{water} \neq \text{earth}$
- Transitive ✗: $S(\text{earth}, \text{water}) \land S(\text{water}, \text{wind}) \not\Rightarrow S(\text{earth}, \text{wind})$
- Partial ✗: Not antisymmetric nor transitive.
- Totality ✗: $\neg(S(\text{earth}, \text{thunder}) \land S(\text{thunder}, \text{earth}))$

c. The relation T on the set of eight trigrams is defined as follows: $T(A, B)$ if A and B have the same number of *yin* lines.

What kind of relation is T, which of the properties of relations does it have?

- Reflexivity ✓: $T(\text{heaven}, \text{heaven})$
- Symmetric ✓: $T(\text{lake}, \text{fire}) \land T(\text{fire}, \text{lake})$
- Antisymmetric ✗: $T(\text{lake}, \text{fire}) \land T(\text{fire}, \text{lake}) \land \text{fire} \neq \text{lake}$
- Transitive ✓: $T(\text{lake}, \text{fire}) \land T(\text{fire}, \text{wind}) \land T(\text{lake}, \text{wind})$
- Partial ✗: Not antisymmetric
- Equivalence ✓: Reflexive, symmetric and transitive
- Totality ✗: Heaven has no relation with any other trigram

7. On the set $A = \mathbb{Z}$, define a relation R by

$$R(a, b) \iff |a| = |b|$$

a. Prove that R is an equivalence relation.

Let $(a, a) \in R$

$|a| = |a|$

Therefore, R is reflexive.

Let $(a, b) \in R$

$(|a| = |b|) \land (|b| = |a|)$

$R(a, b) \land R(b, a)$

Therefore, R is symmetric.

Let $(a, b) \in R, (b, c) \in R$

$|a| = |b|$

$|b| = |c|$

$|a| = |c|$

Therefore, R is transitive.

Hence, R is an equivalence relation.

b. Describe all equivalence classes of R.

As we're dealing with the integers, containing both positive and negative whole numbers, we can group our equivalence classes by whether they're equivalent to a positive integer.

Examples:

$$[1] = \{-1, 1\}$$

$$[2] = \{-2, 2\}$$

$$[67] = \{-67, 67\}$$

This can be generalised by:

$$[x] = \{-x, x\}, x \in \mathbb{Z}^+$$

In the case of zero:

$$[0] = \{0\}$$